Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Prof. Tobin Filleter	Email	filleter@mie.utoronto.ca
Institute Name	NanoMechanics and Materials Lab – University of Toronto			
Address	27 King's College Cir, Toronto, ON M5S 1A1			

2. Technology Overview

Title of Proposed Technology	Multiscale Additive Manufacturing Platform for Nano-Architected Materials Design and Scale-Up			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum (√)			
eIndustrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical / Materials		Electrical / Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Nano/Micro Mechanical Systems	System Integration Technologies	Semiconductor Devices and Systems	MEMS Element
Concept of Proposed Technology	We propose platform technology [1,2] for the design and manufacture of nano-to-centimeter-scale architected carbon/ceramic materials that combine aerogel-like ultra-low density with engineering-grade robustness, including high strength, stiffness, toughness, energy absorption, and functionalities (e.g., thermal). This enables targeting applications that span multiple length scales, including small-scale components for Micro-Electro-Mechanical Systems (MEMS) and quantum computing systems, small- and large-scale protective packaging/coatings, and structural shields for sensitive semiconductor/electronics tooling in aerospace/space applications operating under extreme conditions (e.g., impact, thermal and radiation extremes), biomedical scaffolds and implants, and integrated mechanical-thermal integration components for electronics, as shown in Figure 1 below. In contrast to conventional nano-architected materials approaches, which are largely limited to			

	laboratory-scale demonstrations, this platform integrates generative-AI-informed architectural design, multiscale additive manufacturing, and validated process-property qualification to enable reliable scale-up and application-ready performance. The platform will deliver large-scale demonstrators and scalable manufacturing recipes, while establishing the fundamental manufacturing-microstructure-structure- property relationships required for systematic optimization and translation. Although initially validated through mechanical performance, the same manufacturing and design framework is broadly enabling and can be readily extended to architected materials and devices for applications beyond structural mechanics, with strong engineering versatility across multiple material systems (e.g., glass, metal, polymer) [3] and length scales.
Rationale for Joint R&D Project	Industries increasingly demand material performance that exceeds the capabilities of conventional materials for material-efficient applications such as lightweight protection, impact mitigation, vibration tolerance, and operation in extreme environments. However, existing material solutions face a fundamental trade-off: high strength and stiffness are typically accompanied by high density, while lightweight materials often lack the strength, toughness, or reliability required for demanding service conditions. Recent advances in nano-scale additive manufacturing provide a promising pathway to overcome this limitation by harnessing the exceptional intrinsic properties of nanomaterials through architectural design, while scaling them toward practical form factors. For example, nano-architected carbon materials have achieved carbon-steel– and titanium-alloy–level strength at foam-like densities, while remaining lightweight and delivering exceptional energy absorption—surpassing conventional materials, including diamond-like systems, under both quasi-static and dynamic loading conditions [11,12,13,14]. Despite these advances, meaningful industrial translation requires far more than laboratory-scale printing. Key challenges include scalable manufacturing routes, robust packaging and integration as coatings or structural components, and application-specific qualification of mechanical performance, impact resistance, thermal stability, and long-term reliability. A joint R&D project is uniquely positioned to address these challenges by leveraging complementary expertise from both partners. The NanoMechanics and Materials Lab at the University of Toronto contributes strengths in nano-architected material design, multiscale mechanical characterization, and fundamental structure-property relations, alongside the optimization of printing parameters for scale-up. These capabilities are synergistically combined with Korea's strengths in application-driven qualification, prototype integration, and validation under relevant service conditions. Together, this collaboration is essential to achieve the project's objectives: designing high-performance architectures, establishing a validated multiscale manufacturing workflow, and delivering application-ready demonstrators, such as lightweight protective layers for semiconductor devices and extreme environment support structures, for critical devices.

(max 10 pages)

Key Words	1. Nano-Architected Metamaterials Design 2. Nano-Additive Manufacturing 3. Scaling-Up Manufacturing 4. Generative AI-Informed 3D Design 5. High Throughput Design, Manufacturing, and Framework

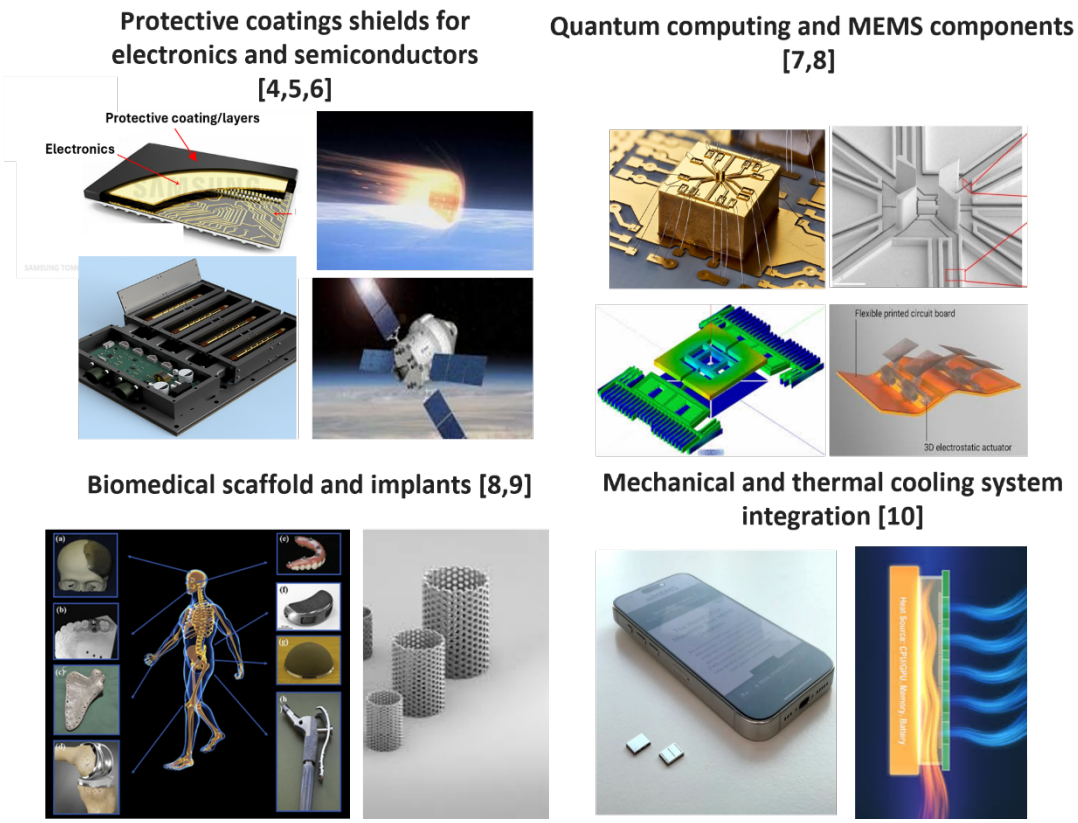

Figure 1: Example of potential applications including small-scale components for Micro-Electro-Mechanical Systems (MEMS) and quantum computing systems, protective packaging/coatings, and structural shields for sensitive semiconductor/electronics tooling in aerospace/space applications operating under extreme conditions, biomedical scaffolds and implants, and integrated mechanical-thermal integration components for electronics.

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	The overall objective of this project is to develop a scalable design-and-manufacturing workflow for nano-architected metamaterials that deliver exceptional material efficiency and mechanical performance in extreme environments, including the protection and support of sensitive devices and electronics in semiconductors, space, and defense applications. This objective will be achieved through the following sub-objectives: 1. **Develop a generative-AI-informed architecture design library** that enables systematic and predictive tuning of strength, damage tolerance/toughness, and energy absorption across multiple length scales. 2. **Establish and validate a multiscale manufacturing workflow**, spanning micro- and nano-scale printing to centimeter-scale components, including the development of robust processing recipes and comprehensive mechanical property validation for carbon and ceramic architected materials. 3. **Demonstrate application-ready prototypes**, including protective coatings and lightweight structural supports, and validate their performance under relevant mechanical loading and environmental conditions representative of extreme-service applications.			
R&D Target Specifications	**Printing Speed Throughput:** 1.2 x 10^8 voxels/second [15] **Printing Dimensions and Time Duration:** 30 x 30 x 0.32 mm^3 in less than two hours [15] **Density / Relative Density:** With foam-like lightweight (0.1 – 1.0 g/cm^3) and carbon-steel-like strength (300 – 800 MPa) [12] **Energy Adsorption:** Energy absorption beyond polymethyl methacrylate and Kevlar (specific energy absorption >30 kJ/kg) [15] **Coating Implementation / Protective Layers:** Conformal architected layer thickness 50 – 500 micrometers with adhesion suitable for handling and processing **Extreme Environment Option:** Retain mechanical integrity after thermal exposure (carbon: up to ~800°C inert; ceramic-converted: >1200°C target, application-dependent)			
R&D Plan	The schematic of the workflow is shown in Figure 2. This plan will be primarily carried out by Ph.D. candidates Jong Hyun Sa, Shuo Yang, and Songlin Duan in the NanoMechanics and Materials Lab. Jong Hyun Sa is affiliated with the Association of Korean-Canadian Scientists and Engineers (AKCSE) and serves as a Vice President of the Young Generation and Professionals National Committee of AKCSE.			

(max 10 pages)

Phase I. Generative AI-Informed, Driven Structure Design (Months 1 – 8):

Generative AI-guided 3D Architecture design:

- Build a state-of-the-art generative design pipeline that can proposes 3D lattice architectures conditioned on target properties (low-density + energy absorption).

Training Data Generation + Simulation Loop:

- A key requirement for a successful AI generative model is a comprehensive database linking architectural topology to mechanical response.
- Generate a topology dataset by hybridizing the existing design space with newly generated geometries.
- Perform high-throughput finite element screening using Abaqus/ (quasi-static) and/or Abaqus/Explicit to extract a mechanical properties dataset.

Manufacturing Feasibility:

- Fabricate selected top candidates using a nano additive manufacturing technique, such as Nanoscribe, to verify geometric fidelity.
- Use Scanning Electron Microscopy (SEM) characterization to measure feature dimensions and establish the selected printing parameters as the baseline process window.

Milestone:

- Developed 5000 – 8000 architectures training datasets.
- Well-trained Generative AI framework.

Phase II. Process Parameters Mapping, Mechanical Assessment, and Optimization Loop (Months 9 – 12):

Printing Parameter Sweep + Mechanical Assessment:

- Fabricate selected architectures using a systematic sweep of printing parameters and perform repeatable mechanical testing before and after conversion (e.g., quasi-static compression).
- Optimize the printing parameters toward qualification metrics (e.g., yield strength).

Coupled Optimization with Pyrolysis / Conversion Conditions:

- Optimize pyrolysis parameters to maximize geometric fidelity, strength, and energy adsorption.
- Build a process-microstructure map linking conversion conditions to the final architecture/microstructure and resulting performance to identify optimal process windows.

Milestone:

- Post processing and pyrolysis / conversion study
- Processing → Microstructure → Properties

Phase III. Scaling Up Parameters Mapping, Characterization, Mechanical Assessment and Optimization Loop (Months 13 – 20):

Scale-Up Strategy + Printing Parameter Mapping (Millimeter to Centimeter Footprints):

- Develop a larger-area fabrication approach to enable printing of millimeter to centimeter scale structures.
- Systematically map and sweep scale-sensitive printing parameters (e.g., stitching, exposure uniformity) to establish a robust scale-up process window.
- Map and optimize pyrolysis / conversion parameters and boundary condition effects (e.g., heating profile) on shrinkage, distortion, and defect formation on final carbon / ceramic structures.

Multi-Length-Scale Characterization and Mechanical Assessment (Geometry → Structural Defects → Property):

- Quantify geometric fidelity and defect populations in both as printed and pyrolyzed parts using SEM for micro-features and X-ray micro-CT for millimeter to centimeter internal integrity.
- Correlate defect statistics and geometry retention with processing conditions to identify dominant scale-up failure modes and mitigation strategies.
- Perform mechanical testing from micro- to cm-scale to quantify property retention (e.g., strength and energy absorption) and assess repeatability across batches.
- Use results to update process windows and feed scale-up constraints back into the design / optimization pipeline.

Milestone:

- Scale-up fabrication (Length and width (1-10cm), height 0.1 -2cm) and mechanical validation

Phase IV. Critical Qualification Testing and Application Prototypes; Validation Under Partner-Relevant Conditions (Months 21 – 36):

Qualification Testing and Application Prototypes

- Prototype types – Protective coatings
- Conditions – Impact, vibration, thermal cycling/aging
- Manufacturability and cost/throughput assessment
- IP and commercialization plan

Milestone:

- Prototype demonstrators and joint technical package for pilot adoption

(max 10 pages)

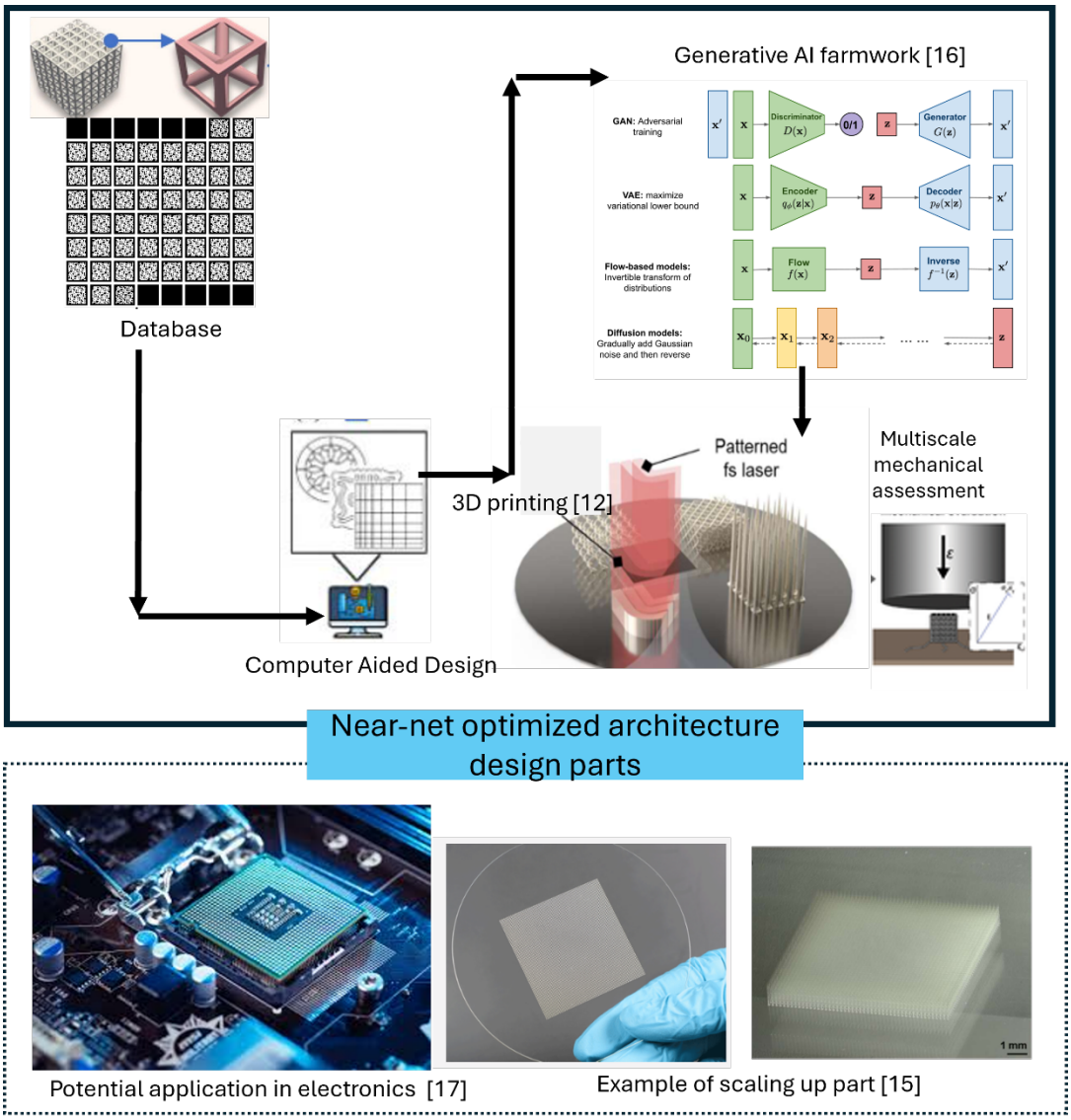

Figure 2: General workflow for optimizing the 3D printing of high-performance nano-architected parts through high-throughput characterization and qualification.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**NanoMechanics and Materials Lab (University of Toronto) brings core expertise in small-scale fabrication and mechanical validation, with representative examples shown in Figure 3 below. They will:** ▪ Lead generative AI-informed architecture design, simulation, and design-for-manufacturing rules, in collaboration with KAIST in Korea [12]. ▪ Execute micro / nano fabrication and post-processing [13]. ▪ Perform microstructure characterization and mechanical testing (quasi-static, impact where applicable) [12]. ▪ Develop a larger-area fabrication approach recipes for carbon and ceramic arrectores (cm).

(max 10 pages)

Expected R&R of Korean SME Partner	**SME Partners:** ▪ Co-develop scalable manufacturing routes (tooling, larger-area additive manufacturing) and quality control. ▪ Integrate architected materials as coatings or components in SME-relevant hardware; define design constraints and interfaces. ▪ Conduct application-specific validation (reliability, environmental qualification, customer relevant standards). ▪ Support pilot production planning, cost-throughput analysis and commercialization activities

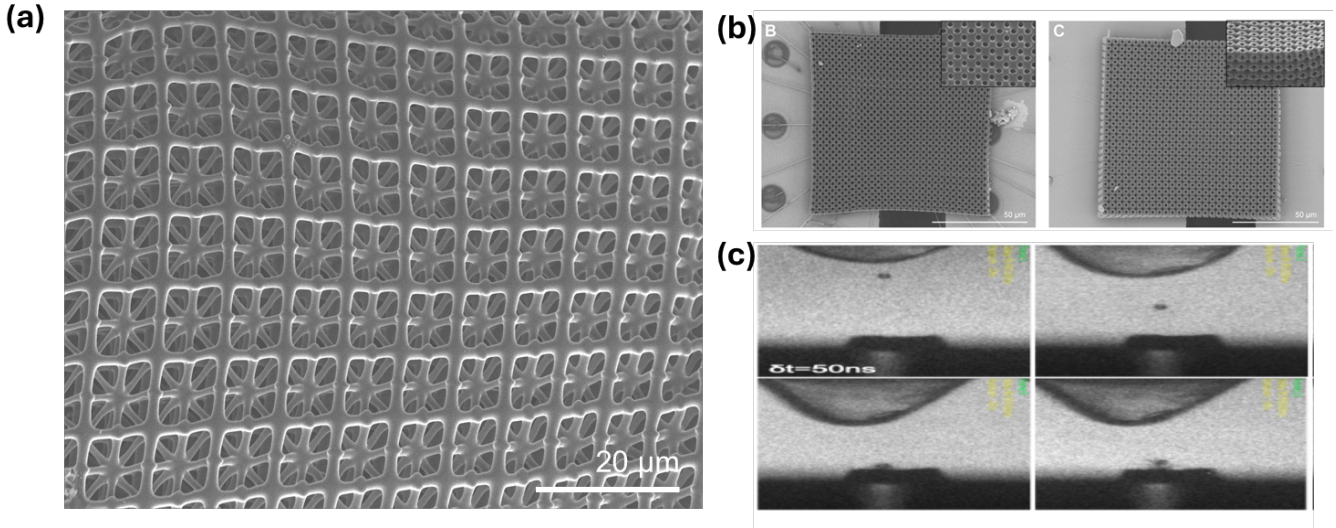

Figure 3: Example carbon nanolattice from our previous work. (a) Screenshot of a portion of a ~3 mm³ nanolattice containing ~18.6-million-unit cells; the unit-cell geometry is AI-informed [12]. This work was conducted in collaboration with the KAIST group in Korea. (b) Ongoing experiments quantifying impact resistance and energy absorption, conducted in collaboration with the Portela Group at the Massachusetts Institute of Technology (MIT).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes through the Joint Program:** 1. This project will deliver a scalable, AI-informed design-and-manufacturing workflow for nano-architected metamaterials, enabling translation from micro to mm-cm with strong process control and high property retention. It will also produce validated prototypes that combine aerogel-like low density with high strength, toughness, and energy absorption, together with practical guidelines for integrating these materials as protective coatings for sensitive electronics exposed to extreme environments. 2. The project will directly address key knowledge gaps in the manufacturing-microstructure-structure-property-performance relationships that currently limit industrial translation (namely, how to reliably design, print, and post-process nano-scale carbon / ceramic architectures into robust, large-area components with retained property, and how to qualify their performance for deployment under extreme operating conditions. **Industrial and Societal Impact:**

(max 10 pages)

	1. This research will pave the way for scaling these nanoarchitectures into high-performance coatings and structural components for critical devices, strengthening Korea's and Canada's leadership in high-value manufacturing and materials innovations, setting both countries as a core for cutting-edge manufacturing, ultimately, attracting investment and creating high-quality jobs. 2. Most importantly, beyond the mechanical demonstrations, the process recipes, design rules, and device concepts developed here are broadly transferable across multiple technology sectors, enabling nano-architected materials/composites to deliver step-change performance from the nano-scale to practical (cm-scale) form. Other potential applications include, but are not limited to, (i) biomedical implants and scaffolds that combine cm-scale geometry with nano-engineered surfaces to improve tissue integration and fatigue life; (ii) energy-device architectures (e.g., battery with integrated mechanical thermal system or catalyst scaffolds) that couple high surface area with mechanical robustness; (iii) vibration / acoustic metamaterials for lightweight damping and noise control; (iv) and filtration or membrane supports capable of operating under harsh thermal and chemical conditions. **3.** Ultimately, this research directly supports Korea's and Canada's strategic priorities in technological innovation, national security, and economic growth, and aligns with national initiatives in both countries – including Korea's National Growth Fund worth 140 billion CAD in advanced industrial technology R&D and manufacturing scale-up, and Canada's federal investment of up to 230 million CAD in the Next Generation Manufacturing Supercluster (NGen), matched by the private sector.

6. References

[1] Jinje, Safa. "Strong as Steel, Light as Foam: Machine Learning and Nano-3D Printing Produce Breakthrough High-Performance, Nano-Architected Materials." U of T Engineering News. January 23, 2025. Accessed January 23, 2026. https://news.engineering.utoronto.ca/strong-as-steel-light-as-foam-machine-learning-and-nano-3d-printing-produce-breakthrough-high-performance-nano-architected-materials/

[2] PR Office. "Ultralight Advanced Material Developed by KAIST and U of Toronto." KAIST News Center. February 18, 2025. Accessed January 23, 2026.

[3] Saccone, M.A., Gallivan, R.A., Narita, K. et al. Additive manufacturing of micro-architected metals via hydrogel infusion. Nature 612, 685–690 (2022).

[4] European Space Agency. "Air and Water." ESA. Accessed January 23, 2026. https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Orion/Air_and_water.

[5] National Aeronautics and Space Administration. "Feel the Heat Activity." Last modified February 14, 2024. Accessed January 23, 2026.

[6] CNN. "Opinion Article about Orion and a Mars Mission." December 3, 2014. https://www.cnn.com/2014/12/03/opinion/bolden-orion-mars-mission. Accessed January 23, 2026.

[7] Xu, S., Xia, X., Yu, Q. et al. 3D-printed micro ion trap technology for quantum information applications. Nature 645, 362–368 (2025).

[8] Nanoscribe. "Micromirror Control in Flexible Microsystems Using 3D Printed MEMS." July 16, 2024. Accessed January 23, 2026.

[9] Vignesh, Margabandu, G. Ranjith Kumar, Mathiyazhagan Sathishkumar, Mano Manikandan, G. Rajyalakshmi, Radhakrishnan Ramanujam, and N. Arivazhagan. "Development of biomedical implants through additive manufacturing: a review." Journal of Materials Engineering and Performance 30, no. 7 (2021): 4735-4744.

[10] Sparrow, Mark. "Electronics Will Soon Be Cooled by Tiny Fans Built Into Chips." Forbes, August 20, 2024. https://www.forbes.com/sites/marksparrow/2024/08/20/electronics-will-soon-be-cooled-by-tiny-fans-built-into-chips/. Accessed January 23, 2026.

(max 10 pages)

[11] Bauer, J., Schroer, A., Schwaiger, R. et al. Approaching theoretical strength in glassy carbon nanolattices. Nature Material. 15, 438–443 (2016).

[12] Serles, Peter, Jinwook Yeo, Michel Haché, Pedro Guerra Demingos, Jonathan Kong, Pascal Kiefer, Somayajulu Dhulipala et al. "Ultrahigh specific strength by Bayesian optimization of carbon nanolattices." Advanced Materials 37, no. 14 (2025): 2410651.

[13] Serles, Peter, Michel Haché, Jason Tam, Alianna Maguire, Tao Li, Guorui Wang, Keith Sebastian et al. "Mechanically robust pyrolyzed carbon produced by two photon polymerization." Carbon 201 (2023): 161-169.

[14] Portela, Carlos M., Bryce W. Edwards, David Veysset, Yuchen Sun, Keith A. Nelson, Dennis M. Kochmann, and Julia R. Greer. "Supersonic impact resilience of nanoarchitected carbon." Nature Materials 20, no. 11 (2021): 1491-1497.

[15] Gu, S., Mao, C., Guell Izard, A. et al. 3D nanolithography with metalens arrays and spatially adaptive illumination. Nature 648, 591–599 (2025).

[16] Weng, Lilian. (Jul 2021). What are diffusion models? Lil'Log.

[17] Liberto, Daniel. "10 Biggest Semiconductor Companies." *Investopedia*. Updated December 2, 2025. Accessed January 23, 2026.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization